Congestion-Aware Joint Ambulance Routing and Selective Traffic Signal Priority: A Reproducible Simulation Study

Abstract. Ambulance signal priority couples emergency travel time with the traffic displaced by each intervention. This study develops a joint routing and selective-priority controller that combines finite route-action search, queue-based intervention scoring, bounded signal requests, and observed-progress escalation. Common-cohort accounting measures ordinary-vehicle residence, spatial occupancy exposure, and individual residence increases, retaining source waiting and unfinished journeys. Evaluation comprises 14,400 mechanically corrected mesoscopic policy runs and 1,728 held-out primary microscopic runs in Eclipse SUMO. Balanced mean restricted ambulance duration was 401.57 s with routing alone, 211.19 s with aggressive joint control, and 222.29 s with selective joint control. The selective-minus-aggressive difference was 11.10 s (95% seed-block bootstrap interval: 9.47-12.94 s), with spatial exposure reduced by 3.84 vehicle-hours (signed interval: -4.73 to -2.95). Both intervals satisfy the evaluated aggregate criterion of a 20 s mean-duration allowance and lower spatial exposure. Progress escalation occurred in 51.6% of selective missions; 300 s exceedance increased by 2.55 percentage points, and two journeys exceeded 30 min. The paired experiments quantify a reproducible time-congestion trade-off and identify high-load reliability as the next control-design priority.

Keywords. Ambulance routing; traffic signal priority; congestion aftereffects; microscopic simulation; finite-horizon accounting; paired experiments.

I. Introduction

Signal priority changes the traffic conditions through which an ambulance travels. Extending a green or advancing a phase clears its approach while postponing service on conflicting approaches. The resulting queues can outlast the intervention and restrict downstream movement. Route selection and signal control therefore form a coupled decision problem: an intervention changes both the ambulance's immediate passage and the downstream conditions that determine the value of its route.

This coupling requires a controller to compare route-priority combinations quickly enough to act on current observations. Each intervention offers an emergency-time benefit and imposes a traffic burden. Aggressive priority emphasizes the first quantity, while selective control accounts for both. We evaluate ambulance duration and ordinary-traffic aftereffects as separate outcomes, making the operational trade-off explicit and preserving the units needed to interpret each effect.

We develop JSEL, a hybrid selective-aggressive controller that evaluates a finite library of route continuations and priority settings. It restricts candidates to a computed ambulance-time band, ranks eligible choices by predicted burden, and executes requests through minimum-green, clearance, receiving-space, and competing-service constraints. An observed-progress rule escalates an individual mission to aggressive joint control when its accumulated movement deficit exceeds a fixed threshold. This design links economical intervention during routine progress with stronger priority when measured progress deteriorates.

The study asks whether this policy preserves most of the mean ambulance-time benefit of aggressive joint control while reducing ordinary-traffic aftereffects. We distinguish three outcomes: signed change in restricted ordinary-vehicle residence, positive spatial occupancy exposure, and positive individual residence increase. These measures use the same desired-trip cohort and horizon. A vehicle held outside the simulated network remains in the accounting, as does a journey unfinished at the endpoint.

Two computational studies examine this design on a common synthetic district. The mesoscopic experiment verifies admission, storage, and discharge mechanics through corrected reruns. A separately specified microscopic implementation then compares routing alone, sequential routing and priority, aggressive joint control, and JSEL under repeated missions and imperfect observations. Together, these layers connect mechanical verification with a controlled assessment of vehicle-level traffic interactions.

The study integrates joint candidate selection, selective request execution, progress-based escalation, and auditable aftereffect accounting in an executable controller. Accounting identities reconcile population traces with individual event records, while a computed-time eligibility property verifies route-selection logic. Matched scenarios, held-out evaluation seeds, and paired uncertainty estimates connect these implementation checks to the measured ambulance-time and congestion outcomes.

II. Related Work

A. Integrated emergency control

Hosseinzadeh et al. [1] formulate emergency-vehicle-centered model predictive control for multi-intersection networks, combining signal decisions, inlet-flow regulation, and an extension for uncontrolled inflows. Their work provides a basis for coordinated emergency control. The present controller combines finite route-action search, approximate forecasts, bounded requests, and progress-based escalation. This design connects candidate scoring and executable interventions to measured traffic aftereffects. Comparisons across these approaches require common observations, control authority, and computational budgets.

Joint emergency routing and signal control have also been studied through decentralized reinforcement learning and hierarchical road- and lane-level MPC. EMVLight coordinates emergency routing and signal preemption while accounting for ordinary traffic [2]. Dai et al. couple dynamic road/lane routing with preemption in a partially connected environment [3]. These studies motivate the joint formulation evaluated here through matched internal comparators.

Varaiya [4] establishes max-pressure stability results for an idealized store-and-forward traffic network. Gregoire et al. [5] develop capacity-aware backpressure control for finite storage and congestion propagation. Zoabi et al. [6] examine stability under restrictions on decentralized max-pressure operation. These studies motivate explicit treatment of downstream receiving space and executable signal phases. Here, a queue-based burden score ranks prospective priority requests, and a longest-count signal rule supplies a separate sensitivity comparison. Each is specified through the observations and actions it uses.

B. Traffic representation and urban validation

Conservation and sending-receiving constraints underpin the cell transmission model [7]. Microscopic simulation adds individual vehicle dimensions, following, lane changes, and junction connections. SUMO supplies these mechanisms and external control interfaces [8]. With direct signal-state commands, the controlling application specifies the requested transitions [9], [10]. We therefore check population conservation and signal execution explicitly. FHWA guidance further structures empirical model assessment through calibration and alternatives analysis [11], [12], providing the basis for subsequent evaluation on observed corridors.

Urban-scale accumulation and production relationships depend on spatial structure and measurement. Geroliminis and Daganzo [13] report empirical macroscopic fundamental diagrams, and Loder et al. [14] relate urban network properties to capacity. These findings motivate reporting actual queues, receiving space, and spillback alongside nominal demand indices. For heterogeneous traffic, Kanagaraj and Treiber [15] emphasize width-sensitive lateral interactions. The present synthetic experiments represent vehicle heterogeneity through class-specific dimensions and passenger-car-unit factors; extension to Indian corridors calls for calibration of lateral behavior and emergency yielding.

C. Evaluation and contribution

Inference follows the experimental dependence structure. Bootstrap resampling [16] retains ambulances sharing an episode and each seed's complete vector across conditions. Exact binomial intervals [17] use independent episodes for rare-event summaries. Holm adjustment [18] and Monte Carlo permutation methods [19] depend on valid component tests and the relevant exchangeability assumptions. The main analysis reports paired effects and seed-block intervals for the scenario generator, with explicit mission- and episode-level denominators.

The contribution integrates a reproducible hybrid controller, matched experiments at two traffic resolutions, and complementary spatial and traveler-level measures of intervention cost. Each reported effect can be traced from policy decisions and physical execution to the population or event records used in its estimation.

III. Problem Formulation and Traffic Accounting

A. Information and control objective

Let $G=(V,E)$ be a directed road network with legal turning connections, finite road storage, and signal automata. The full state X_k contains physical traffic, source queues, signal stages, and pending requests. At physical step k,

$$X_{k+1}=F(X_k,U_k,Z_k). \tag{1}$$

Here Z_k contains exogenous scenario innovations. A causal controller chooses $U_k=\pi_k(\mathcal{I}_k)$ from commands admissible under its information history $\mathcal{I}_k$. This history includes public geometry and nominal signals, received noisy traffic counts, current ambulance telemetry, and previous commands. It excludes future realized demand and the counterfactual baseline trajectory used later for evaluation.

For a mission dispatched at d_a, let e_a^π be its arrival time, with infinity assigned to nonarrival. Its common-horizon restricted duration is

$$T_a^\pi=min(e_a^\pi,H)-d_a. \tag{2}$$

All scheduled dispatches precede H. Duration includes insertion waiting; completion status is retained separately. Write $\overline{T}^\pi$ for the episode mean. Over an admissible causal policy class Π, the ideal time-first objective is

$$T^*=inf_{\pi\in\Pi}\mathbb{E}[\overline{T}^\pi]. \tag{3}$$

An ideal secondary problem minimizes expected positive spatial exposure among policies within an allowance δ:

$$min_{\pi\in\Pi}\mathbb{E}[A_H^\pi], \qquad \mathbb{E}[\overline{T}^\pi]\le T^*+\delta. \tag{4}$$

Equations (3)-(4) define the time-first objective and secondary congestion allowance. The planner approximates this objective with a finite candidate set and forecast tolerance ε. The empirical assessment separately uses a 20 s allowance for the balanced mean-duration contrast. Thus δ defines the ideal allowance, ε controls snapshot eligibility, and 20 s defines the evaluation criterion. Completion status and 300 s and 600 s exceedance provide complementary reliability outcomes.

B. Restricted ordinary-vehicle residence

Fix an ordinary-vehicle cohort $\mathcal{C}$, desired release times d_i, and an observation interval $[W,H]$. Let e_i^π be the exit time under policy π, or infinity for an unfinished trip. Define

$$I_i^\pi(t)=1\{d_i\le t<e_i^\pi\}, \tag{5}$$

$$N^\pi(t)=\sum_{i\in\mathcal{C}} I_i^\pi(t). \tag{6}$$

A desired but uninserted vehicle counts as present. For $[x]_+=max(x,0)$, define its restricted residence as

$$j_i^\pi=[min(e_i^\pi,H)-max(d_i,W)]_+. \tag{7}$$

Proposition 1. The aggregate residence identity is

$$J_H^\pi = \int_W^H N^\pi(t)\, dt = \sum_{i \in \mathcal{C}} j_i^\pi. \tag{8}$$

Proof. Substitute the finite sum of presence indicators into the integral and interchange summation and integration. Each term equals the length of that vehicle's residence interval intersected with [W, H]. The identity covers source waiting, completed trips, and trips still present at the endpoint.

The signed baseline-relative effect, ΔJ = J(policy) - J(baseline), is measured in vehicle-seconds over a common window. Residence begins at desired release, retaining external source waiting. The accounting follows the same cohort through source queues, physical-network travel, and endpoint censoring. Population integration can therefore be reconciled directly with the individual event ledger.

C. Spatial and individual aftereffects

Fix an exhaustive, policy-invariant spatial partition $\mathcal{B}$. Every present ordinary vehicle belongs to exactly one group, including internal junction regions and external origin queues. Set $z_b^\pi(t) = N_b^\pi(t) - N_b^0(t)$ and define

$$A_H^\pi = \int_W^H \sum_{b \in \mathcal{B}} [z_b^\pi(t)]_+ \, dt, \tag{9}$$

$$B_H^{-,\pi} = \int_W^H \sum_{b \in \mathcal{B}} [-z_b^\pi(t)]_+ \, dt. \tag{10}$$

Proposition 2. For the same cohort, partition, and horizon,

$$\Delta J_H^\pi = A_H^\pi - B_H^{-,\pi}, \tag{11}$$

$$A_H^\pi \geq [\Delta J_H^\pi]_+. \tag{12}$$

Proof. Apply $x = [x]_+ - [-x]_+$ to each group's occupancy difference, sum over the exhaustive partition, and integrate. The second statement follows because negative exposure is nonnegative.

Positive spatial exposure accumulates local occupancy increases before aggregation. It includes moving and stopped vehicles, preserving the location of traffic redistribution. Its dependence on reporting resolution follows from:

$$[\sum_b z_b]_+ \leq \sum_b [z_b]_+. \tag{13}$$

Merging groups weakly reduces positive exposure for fixed trajectories, so comparisons retain a common partition. Extending the horizon weakly increases positive exposure, while signed residence can move in either direction during recovery. The observation window is therefore part of each reported traffic effect.

Matched vehicle identities yield complementary traveler-level measures:

$$D_H^{+,\pi} = \sum_i [j_i^\pi - j_i^0]_+, \tag{14}$$

$$D_H^{-,\pi} = \sum_i [j_i^0 - j_i^\pi]_+. \tag{15}$$

Proposition 3. The individual decomposition satisfies

$$\Delta J_H^\pi = D_H^{+,\pi} - D_H^{-,\pi}, \tag{16}$$

$$D_H^{+,\pi} \geq [\Delta J_H^\pi]_+. \tag{17}$$

Proof. Apply the positive–negative decomposition to each paired residence difference and sum over the common cohort.

Spatial exposure and positive individual residence describe complementary effects. Spatial redistribution can change local occupancies while preserving completion times, and traveler-level increases can coexist with improvements elsewhere. Their relative sizes depend on those trajectories. Both measures dominate the positive network-accumulation area:

$$C_H^{+,\pi} = \int_W^H [N^\pi(t) - N^0(t)]_+ \, dt. \tag{18}$$

For spatial exposure, the bound follows from the aggregation inequality. For individual residence, paired presence indicators share a release time, so each paired difference has a constant sign wherever it is nonzero. Summing positive individual differences before aggregation gives the second bound. Both positive measures lie between zero and policy residence. These relations provide independent accounting checks. Full-trip duration differences are recovered when the window contains both residence intervals, or when an omitted common prefix cancels.

D. A conditional recovery calculation

Consider an initially empty fluid queue with constant arrival rate $\lambda > 0$ and service rate $\mu > \lambda$. The baseline serves arrivals immediately. Removing all service for Δ seconds creates a queue of $\lambda\Delta$ vehicles; after service resumes, it clears at rate $\mu - \lambda$. Consequently,

$$t_{clear} = \frac{\lambda \Delta}{\mu - \lambda}, \tag{19}$$

$$C = \frac{\lambda \Delta^2}{2} + \frac{(\lambda \Delta)^2}{2(\mu - \lambda)} \tag{20}$$

$$C = \frac{\mu \Delta^2}{2} \frac{\rho}{1 - \rho}, \qquad \rho = \frac{\lambda}{\mu}. \tag{21}$$

The isolated-queue calculation motivates an inverse-slack burden term: a fixed interruption creates greater residence as utilization approaches capacity. Nominal design loading provides the controller's ranking input. Simulations evaluate the resulting interventions amid cyclic service, queues, spillback, turning, and repeated priority.

More generally, suppose an absolutely continuous paired accumulation difference $z(t)$ starts at $z(W) = Q > 0$ and satisfies $z'(t) \geq -r(t)$ almost everywhere for a nonnegative integrable depletion bound r. Integration gives

$$[z(W+s)]_+ \geq [Q - \int_0^s r(W+u)\, du]_+. \tag{22}$$

For constant r > 0, the bound gives earliest recovery Q/r and positive residence of at least $Q^2/(2r)$ over an interval covering recovery. For a shorter interval, integration uses the truncated lower envelope. Here r bounds relative depletion, incorporating changes in inflow as well as service; applying the result requires an independently justified depletion bound.

IV. Microscopic Joint Controller

A. Physical model and observations

The microscopic implementation uses Eclipse SUMO 1.28.0 through libsumo. The synthetic district has 16 signalized junctions, eight boundary nodes, and 59 directed links. Link lengths are approximately 260–472 m, with one or two 3.2 m lanes and speeds of 8.33 or 11.11 m/s. Internal connections are retained and turnarounds excluded. The district is shared with the mesoscopic study, but the demand, vehicle dynamics, and controller are separately specified.

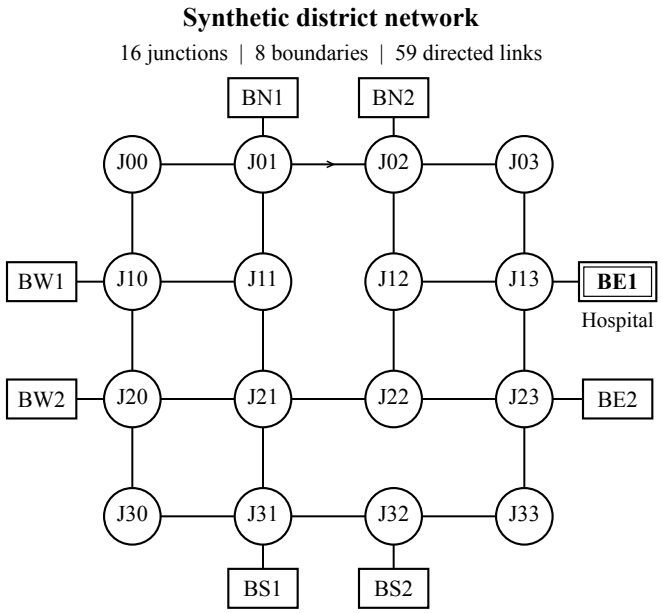


Fig. 1. Simulated road network and hospital destination (BE1).

Ordinary classes are cars, motorcycles, autorickshaws, buses, and goods vehicles, with probabilities 0.53, 0.30, 0.10, 0.04, and 0.03. Their PCU factors are 1, 0.5, 1, 3, and 2.5. The fixture specifies dimensions, gaps, acceleration, deceleration, and headways. SUMO's standard car-following and lane-changing mechanisms use zero imperfection and unit speed factor. Ambulances obey signals and share passenger-class dynamics: length 5 m, width 2 m, gap 2 m, headway 1.3 s, acceleration 2 m/s², and deceleration 4.5 m/s². Their travel benefit arises from routing and signal service. Pedestrian service enters through reserved signal time, and the microscopic plant uses the fixed road network throughout each episode.

Physical dynamics advance every 0.5 s and control updates every second. Episodes start empty, end at 7,200 s, and schedule missions during 1,800-5,400 s. Ordinary demand continues to the endpoint. The initial 1,800 s provides a common initialization interval. Appendix A specifies the demand generator and nominal loading procedure.

Traffic counts are sampled every 5 s and received after 2 s. Additive errors are −2, −1, 0, 1, or 2 vehicles, with probability 3/7 at zero and 1/7 otherwise; negative reports are clipped. At each sampling opportunity, availability changes from available to missing with probability 0.015 and back with probability 0.25. The most recent report is held during dropout. Matched policies share error draws, applied to their own physical counts. Ambulance link and longitudinal position are assumed exact at decision times.

Let $\hat{n}_e$ be the received count, $\overline{\lambda}_e$ the historical arrival rate in vehicles/s, s_e saturation flow in PCU/s, and $\overline{p}$ mean PCU per ordinary vehicle. The queue and discharge proxies are

$$\hat{q}_e = [\hat{n}_e - \overline{\lambda}_e \ L_e / v_e]_+, \tag{23}$$

$$\hat{d}_e = \frac{\hat{q}_e}{max(s_e / \overline{p}, 0.1)}. \tag{24}$$

The discharge floor is 0.1 vehicles/s. Received counts combine moving and queued traffic and retain the latest report during dropout. Equations (23)-(24) map these observations to approximate queue and discharge inputs.

B. Route and action scoring

On entry to a regular link, up to four loop-free downstream continuations to BE1 are generated within 1.5 times the shortest downstream free-flow continuation, excluding the common current-link remainder. Immediate U-turns are excluded. A legal incumbent replaces the fourth candidate when necessary, as an explicit exception to the 1.5 filter. An action $a = (h, \ell)$ specifies trigger look-ahead and requested green hold. Development considered the library {(0, 0), (10, 20), (20, 40), (30, 60)} s and its strong subset {(0, 0), (30, 60)} s.

Let $w_e(\tau)$ be residual nominal signal wait at predicted arrival offset τ. Forecasts start from the current stage and assume subsequent nominal cyclic progression. The legal-switch delay δ_e is zero for the ambulance's green; remaining mandatory green plus 5 s for a conflicting green; remaining amber plus 2 s during amber; remaining clearance during all-red; or the remaining pedestrian reservation. Downstream forecasts apply this current-stage approximation at each candidate arrival offset.

For a nonzero action, the feasibility proxy and receiving condition are

$$[h + \hat{d}_e - \delta_e]_+ + \hat{d}_e + 2 \le \ell, \tag{25}$$

$$\hat{n}_{out} < 0.85 K_{out}. \tag{26}$$

The two seconds are a start-up allowance, and K_{out} is the declared aggregate receiving-storage count. If both checks pass, predicted benefit is

$$b_e = [w_e(\tau) - [\delta_e - \tau]_+]_+. \tag{27}$$

Otherwise it is zero. This observed-headroom check applies to aggressive scoring as well. Actual aggressive requests bypass that admission gate; physical receiving constraints remain binding. The no-priority action always has zero benefit. The displaced-service duration and local burden are

$$D_e = min\{\ell, max(5, h + \hat{d}_e)\}, \tag{28}$$

$$\hat{C}_e = \sum_{j \neq e} (\hat{q}_j D_e + \frac{\overline{\lambda}_j D_e^2}{2 max(1 - X_j, 0.05)}). \tag{29}$$

The five-second duration floor and dimensionless 0.05 slack floor are fixed constants. The sum covers conflicting approaches, and X denotes nominal design loading. The vehicle-second score ranks local interventions while limiting growth near nominal saturation. Admission requires benefit of at least 5 s, burden within the admission threshold, feasible hold duration, and receiving headroom. Realized network exposure is measured separately from simulation trajectories.

A link ending at an unsignalized node has zero signal wait and zero priority burden. For action (0, 0), a signalized link contributes nominal wait without a priority benefit. Otherwise, candidate ETA accumulates remaining free-flow travel on the current link, full free-flow travel on later links, discharge proxies, and residual signal waits less admitted benefit. Candidate burden sums admitted local scores. For the nonempty evaluated domain $\mathcal{C}_k$, set

$$\hat{T}_{k,min} = min_{c \in \mathcal{C}_k} \hat{T}_k(c), \tag{30}$$

$$\mathcal{E}_k = \{c \in \mathcal{C}_k : \hat{T}_k(c) \leq \hat{T}_{k,min} + \varepsilon\}. \tag{31}$$

Initial selection minimizes the tuple (burden, ETA, route identifier, action identifier) over $\mathcal{E}_k$. A 10 s switching preference may retain the incumbent when its best ETA exceeds the proposed ETA by no more than 10 s. In selective mode, retention is allowed only for an incumbent action that remains in the original global eligible set; the least-burden eligible incumbent action is then preferred.

Proposition 4. Every completed selective-mode planning call preserves $\hat{T}_k(c_k) \leq \hat{T}_{k,min} + \varepsilon$, provided each selection and hysteresis branch enforces that same global cap.

Proof. The initial choice belongs to $\mathcal{E}_k$. The only permitted replacement is an incumbent action filtered by the identical cap. Either branch therefore returns an eligible candidate.

Proposition 4 verifies snapshot eligibility under a common computed-time cap. Hysteresis can retain an eligible incumbent with greater predicted burden to preserve route persistence. Paired travel times, traffic accounting, and the escalation experiment evaluate forecast accuracy and realized aftereffects.

C. Request execution and competing missions

A request is considered when remaining-distance free-flow ETA is at most $h + \hat{d}_e$. For JSEL, the benefit and burden gates apply to a new request. Once admitted, a bounded lease keeps it active even if successful service removes its estimated benefit. On the first observed own-green stage, the lease fixes an absolute expiry at the current time plus the nonnegative difference between the requested hold and the already elapsed green. That expiry clock does not pause during interruptions. Release occurs at expiry, 120 s after admission, or loss of observed receiving headroom. Expiry and headroom termination make the ambulance-approach pair nonrenewable. Link departure removes the lease, allowing reconsideration after a physical revisit. The cap applies to discretionary holding; nominal signal timing continues under the common automaton.

Conflicting requests are ordered by scheduled dispatch and identifier, with the oldest eligible mission supplying the junction request. This deterministic arbitration gives each run a reproducible service order. Junctions protect one incoming approach at a time, with 8 s minimum green, 3 s amber, and 2 s all-red. A started 25 s pedestrian opportunity completes as an all-red reservation. The cyclic pointer then synchronizes to the served approach.

A service-age guard gives precedence to an observed occupied approach, or pedestrian opportunity, aged at least 150 s; the most overdue wins. It operates on service opportunities, with realized vehicle waits recorded separately. The longest-count sensitivity rule selects the largest observed approach count at discretionary decisions, providing an alternative base-signal selection mechanism.

D. Observed-progress escalation

Let $\mathcal{L}_a(t)$ be the sequence of completed regular-link traversals for ambulance a, preserving repeated traversals. On its current regular link e, let $x_a(t)$ be longitudinal position. Executed free-flow credit is

$$P_a(t) = \sum_{f \in \mathcal{L}_a(t)} \frac{L_f}{v_f} + \frac{max(0, x_a(t))}{v_e}, \tag{32}$$

$$D_a(t) = [t - d_a - P_a(t)]_+. \tag{33}$$

The current-link term lies outside the completed-link sum. Evaluation occurs at one-second observations on regular links, not before insertion or on internal connectors. Connectors earn no separate free-flow credit; their elapsed time contributes at the next regular-link observation. Insertion waiting similarly enters when the ambulance first becomes observable.

If the progress deficit exceeds γ, the mission's escalation flag is set irreversibly, its selective lease is cleared, and it replans immediately, including mid-link. It then uses aggressive joint scoring and the (30, 60) s request for the remaining mission. Other missions retain their own flags. Receiving constraints, arbitration, clearances, and service-age protection remain common. The threshold acts on executed free-flow credit, including completed detours, and marks the transition from selective eligibility to time-oriented control.

E. Comparator policies and development

S0 uses traffic-aware routing without discretionary priority. SEQ selects routes without anticipating priority, then applies aggressive requests. JFAST anticipates those requests during time-oriented route selection. JSEL uses the

selective rule and progress switch above. SEQ and JFAST use 30 s look-ahead, 60 s holds, and 10 s route hysteresis. All share the physical plant and signal constraints.

Development used seeds 5001–5006 at nominal loads 0.7 and 0.9 with four missions. The initial grid contained 12 unguarded selective configurations and two reference controllers, totaling 168 runs. None met the development margin of 20 s mean duration above JFAST. A second grid added progress escalation and tested nine combinations of B_{adm} in {600, 2400, 10000} vehicle-seconds and γ in {5, 10, 20} s, plus the two references, totaling 132 runs. The strong library and $\varepsilon = 0$ were used.

Among configurations meeting the development margin, selection minimized mean positive spatial exposure. The chosen values were $B_{adm} = 10000$ vehicle-seconds, $\gamma = 20$ s, and $\varepsilon = 0$. Development differences from JFAST were +17.5625 s in duration and −3810.875 vehicle-seconds in exposure. The 2400 threshold tied on these scores; deterministic label order selected 10000. Adaptive development established the final configuration; the implementation and selected parameters were then frozen for evaluation on separate seeds.

The one-second execution order is: assimilate available observations; update mission progress and escalation; replan on regular-link entry or escalation; maintain admitted leases and generate new requests; arbitrate conflicts; apply the signal automata; take the left-endpoint census before each physical half-step; advance the plant; and collect entry/exit events for reconciliation. Baseline trajectories are used only after paired runs. The detailed scoring and execution procedure is provided in Appendix B.

V. Experimental Design

A. Main evaluation and accounting

The main microscopic grid combines 48 seeds, three nominal loads {0.60, 0.80, 0.95}, three mission doses {1, 4, 10}, and four policies: 1,728 policy runs in 432 matched scenario groups. Each group shares ordinary desired releases, vehicle classes, fixed routes, and mission schedules. Seed identifiers recur across load-dose conditions. Aggregate inference therefore resamples complete seed blocks, preserving both within-scenario policy pairing and dependence across the nine conditions.

Ordinary residence is measured from $W = 1800$ to $H = 7200$ s. The exhaustive partition has 59 regular links, one aggregate internal-junction group, and eight origin queues of desired but uninserted vehicles. Ambulances are excluded. At each physical step, counts must equal desired ordinary releases minus ordinary completions. Left-endpoint integration is reconciled against the individual event ledger. Positive spatial exposure includes moving as well as stopped vehicles. Positive individual residence is a secondary endpoint; selection uses spatial exposure.

Execution logs track collisions, attempted teleports, signal intervals, route changes, requests, lease expiry, and observation availability. Teleportation is disabled. Execution exceptions are retained as failures. The analysis reports common-horizon restricted ambulance means, completion counts, a fixed 600 s restriction, and 300 s and 600 s exceedance. These experimental summaries describe distinct aspects of the journey-duration distribution.

B. Estimation and uncertainty

For seed r and condition s, the primary paired differences are $\Delta T_{rs} = \overline{T}_{rs}^{JSEL} - \overline{T}_{rs}^{JFAST}$ and $\Delta A_{rs} = A_{H,rs}^{JSEL} - A_{H,rs}^{JFAST}$. With nine equally weighted conditions,

$$\theta_T = \frac{1}{9}\sum_s \mathbb{E}_r[\Delta T_{rs}], \quad \theta_A = \frac{1}{9}\sum_s \mathbb{E}_r[\Delta A_{rs}]. \quad (34)$$

The joint assessment requires a balanced mean-duration contrast at most 20 s and a negative spatial-exposure contrast. The allowance originates in adaptive development. The retained protocol records the controller freeze and held-out design; the analysis applies this development-derived criterion to the paired evaluation.

Two-sided 95% percentile-bootstrap intervals use 10,000 resamples. Aggregate inference first averages each seed's paired differences over nine conditions, then resamples complete seed vectors. Within-condition intervals resample 48 paired seeds. Upper interval endpoints assess the directional mean trade-off. Condition-specific intervals describe marginal uncertainty for each cell. Completion and exceedance outcomes characterize mission reliability separately.

Equal condition weighting gives the one-mission and ten-mission settings the same influence on the aggregate estimate. The resulting estimand describes the balanced synthetic grid. Sampling intervals quantify variability across the generator's finite seed sample. Positive traffic outcomes are first computed relative to each scenario's S0 baseline, then contrasted between JSEL and JFAST, preserving the common reference used for both policies.

C. Secondary experiments

Four sensitivity designs use seeds 0-11 at loads 0.60 and 0.95 with four missions: a 0.25 s physical step (72 runs), removal of progress escalation with its baseline (48), longest-count base signals (72), and noiseless counts sampled each second with zero receipt delay and retained dropout (72). The total is 264 runs. Physical-step sensitivity retains one-second signal control and the desired-release grid. SUMO's default vehicle decision interval follows the physical step [20], so this experiment jointly varies physical integration and driver-update timing.

The observation experiment jointly varies noise, sampling, receipt delay, and dropout duration in physical time, since availability transitions occur more frequently. It retains the count-to-queue proxy. Physical-step, base-signal, and observation experiments report changes in the JSEL-minus-JFAST contrasts. The escalation ablation directly contrasts unguarded and guarded JSEL. Each interval is interpreted with its direction and precision, including the range of effects compatible with an interval spanning zero.

TABLE I
Experimental inventory

Experiment	Seeds[1]	Runs
Corrected mesoscopic	200	14,400
Initial development	6	168
Escalation development	6	132
Primary microscopic	48	1,728
Secondary microscopic	12	264

[1] Seeds per condition; the same seed identifiers recur across conditions. Development is excluded from primary inference.

VI. Mesoscopic Verification and Reanalysis

The mesoscopic study supplies a complementary check on modeling mechanics. Its 16-junction district uses finite PCU storage, pooled FIFO approach queues, discharge credit, and nominal signals. One or two deque heads may discharge according to lane count, but lane assignment and lateral movement are not physically resolved. Exit links are unbounded free-flow transport delays; vehicles remain in the completion ledger until exit. Appendix C records the demand, observation, incident, and yielding assumptions.

The six policies are M0 (no priority), M1 (limited extension/truncation), M2 (phase skipping and holding), MQ (queue-triggered longer holds), MC (service-deficit compensation), and MR (queue-aware rerouting). They correspond to archived labels P0, P1, P2, P3, P3C, and P4. They are distinct from the microscopic policy family. The evaluation spans six nominal loads, two yielding settings, six policies, and 200 seeds per cell, totaling 14,400 runs.

Source inspection identified three ambulance exceptions: admission to full storage, crossing with insufficient or unconsumed discharge credit, and insertion 20 m downstream of source admission. Adversarial checks reproduced full-link over-admission and zero-capacity discharge. A blocked-link test recorded 348 s of unfinished stop-line waiting while maximum completed wait stayed zero. This established the need to retain ongoing waits and start travel accounting at desired dispatch.

The corrected implementation inserts the ambulance through the source FIFO at position zero, enforces strict storage, and requires the same PCU discharge accounting as ordinary vehicles. Travel time starts at desired dispatch. Policy logic, nominal signal design, ordinary-demand generation, and the remaining model approximations are retained. Eighteen focused and stress checks passed before all 14,400 corrected runs were completed.

Independent reruns of six policies for seeds 0, 37, and 199 in every original load-yield cell produced 216 verification runs. Checked outputs agreed within absolute tolerance 10^{-9} and relative tolerance 10^{-12}. All 72 summary rows and 48 contrast rows were reconstructed to reported precision. This establishes numerical reproducibility for the checked archive outputs and calculations. Archived sign-flip and Holm calculations are also reproducible; the present effect analysis uses paired bootstrap intervals, preserving the dependence structure of the scenario design.

Without abstract yielding, corrected M2 mean ambulance duration was 204.17 s versus M0's 357.61 s at load 0.60, 252.81 versus 433.35 s at 0.80, and 312.89 versus 478.03 s at 0.90. Across the six loading targets, M2 reduced the mean by approximately 34–43%. Its M0-relative ordinary-residence effect rose from −0.025 vehicle-hours [−0.116, 0.065] at load 0.60 to 1.349 [0.857, 1.873] at 0.90 and 1.663 [1.108, 2.259] at 0.95. Brackets denote 95% paired bootstrap intervals.

Mechanical correction changed both baseline and intervention trajectories. At load 0.90, M2 duration rose from 291.57 to 312.89 s and its baseline-relative residence effect from 0.964 to 1.349 vehicle-hours. The paired change in the M2-minus-M0 effect was +0.384 [0.125, 0.646] vehicle-hours, comparing both implementations against their matching baseline.

At load 0.80, M2 reduced duration relative to M1 by 86.49 s [77.17, 95.88], with ordinary residence +0.088 [-0.176, 0.350] vehicle-hours. At 0.90, MQ-minus-M2 was +1.49 [-4.38, 7.48] s and +0.438 [0.141, 0.754] vehicle-hours, jointly reflecting trigger and hold changes. MR-minus-MC was -10.15 [-18.39, -3.50] s and -0.261 [-0.526, -0.031] vehicle-hours.

Tail outcomes varied with load. Under M2, fractions exceeding 300 s were 0%, 22.5%, 41.0%, and 42.5% at loads 0.60, 0.80, 0.90, and 0.95. For zero exceedances among 200 independent single-mission scenarios, the two-sided 95% exact binomial upper endpoint is about 1.83%. These results connect stronger mean travel benefits with a continuing need to examine high-load journey reliability.

The isolated fluid calculation was independently checked at service 0.5 vehicles/s and a 10 s interruption. Additional queue residence was 25.00, 58.33, 141.67, and 225.00 vehicle-seconds at utilization 0.50, 0.70, 0.85, and 0.90. Piecewise-linear integration differed by less than 3×10^{-12} vehicle-seconds, and a deterministic packet reference converged under refinement. These checks establish the calculation's numerical consistency and illustrate the increasing residence cost of interruptions as service slack decreases.

VII. Microscopic Results

A. Verification and balanced outcomes

All 1,728 primary runs completed with matched scenario schedules and zero maximum census and individual-residence discrepancies. Monitoring recorded zero collisions, teleport starts, or completed signal-timing violations across 19,299,062 intervals. Maximum service age was 258 s. These diagnostics verify monitored simulation mechanics and characterize executable service following the 150 s precedence trigger.

The batch contains 8,640 mission evaluations across policy replays, with up to four ambulances active simultaneously. At the endpoint, 82-852 ordinary vehicles remained unfinished across runs; the largest external source queue held 49 vehicles. Both unfinished trips and external waiting remain in common-cohort residence accounting.

TABLE II
Balanced microscopic means

Policy	Time (s)	A (veh-h)	D^+ (veh-h)	ΔJ (veh-h)
S0	401.57	0.00	0.00	0.00
SEQ	214.68	65.96	22.97	1.86
JFAST	211.19	66.50	23.93	3.89
JSEL	222.29	62.66	22.29	2.63

Balanced mean restricted ambulance durations were 401.57 s (S0), 214.68 s (SEQ), 211.19 s (JFAST), and 222.29 s (JSEL). JSEL-minus-S0 was -179.28 s [-189.46, -169.57], a 44.6% reduction in the balanced simulation means. Anticipating priority in route selection produced a further JFAST-minus-SEQ benefit of -3.49 s [-4.87, -1.97]. This contrast quantifies the value of joint selection within the common candidate library and forecast model.

JSEL-minus-JFAST was +11.10 s [9.47, 12.94] in duration and -3.84 [-4.73, -2.95] vehicle-hours in positive spatial exposure. Both upper endpoints satisfy the aggregate assessment criterion. Positive individual baseline-relative residence changed by -1.64 [-2.14, -1.16] vehicle-hours, and signed residence by -1.26 [-1.94, -0.63]. Each contrast compares traffic outcomes first measured against the matching S0 episode, maintaining a common reference across the two joint controllers.

Relative to S0, JSEL added 2.63 vehicle-hours of signed ordinary residence. Its improvement over aggressive control therefore represents a reduction in the ordinary-traffic cost associated with the ambulance-time benefit.

B. Escalation and reliability

JSEL escalated 1,114 of 2,160 scheduled missions (51.6%) to aggressive control. Its mean-duration performance reflects the combination of selective admission and progress-triggered escalation. The ablation below measures the contribution of that escalation mechanism directly.

Condition-specific estimates show greater uncertainty at high load. At load 0.95 with one mission, JSEL-minus-JFAST was +17.80 [9.74, 27.17] s and -1.28 [-3.04, 0.52] vehicle-hours of spatial exposure. At the same load with four missions, duration increased by 16.17 [10.83, 22.84] s. These intervals place the aggregate finding alongside the range of effects in each operating condition; Appendix D gives the full results.

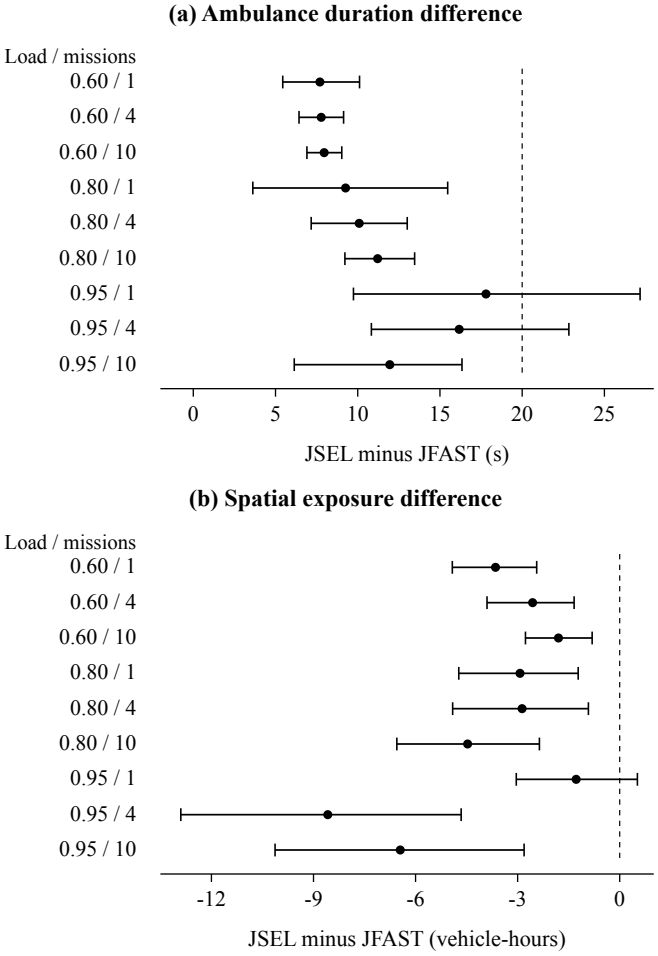


Fig. 2. Paired JSEL-minus-JFAST effects by nominal load and scheduled missions. Bars are marginal 95% seed-bootstrap intervals from 48 paired seeds. The dashed 20 s line is the research tolerance. Spatial exposure is separately measured against the same S0 baseline.

S0 and SEQ each left one of 2,160 missions unfinished in the same seed-2, load-0.95, ten-mission scenario, observed for 2,579 s. JFAST and JSEL completed all missions. Counts exceeding 600 s, including unfinished trips, were 106, 4, 5, and 2 for S0, SEQ, JFAST, and JSEL. Mission counts summarize correlated journeys across heterogeneous conditions. Rare-event uncertainty uses independent episodes within a fixed condition: even zero late episodes among 48 yields a two-sided 95% binomial upper endpoint of about 7.4%.

The two JSEL journeys above 600 s lasted 1,812.5 and 2,050 s, both at load 0.95 in seed 2 with four and ten missions. Insertion delays were 0.5 s. In the first journey, escalation at model time 3,476 s preceded a 1,589 s gap between regular-link entries. At that gap's start, predicted remaining time was 276.8 s against 1,718.5 s realized. The trace locates prolonged delay within the physical network and quantifies the forecast discrepancy that persisted after escalation.

Following the escalation-triggered replan, subsequent replanning occurs at regular-link entry. The long interval between entries explains why an outdated continuation can persist during a stalled mission. The trace identifies the interaction between forecast error and the controller's

link-entry update schedule in this high-load episode, giving the observed journey a concrete control-level interpretation.

The balanced fraction exceeding 300 s was 2.55 percentage points higher under JSEL than JFAST [1.42, 3.77]. At 600 s, the difference was approximately -0.069 percentage points [-0.185, 0.000], based on few extreme events. These threshold-specific results characterize the shape of the reliability trade-off. Fixed 600 s restricted means were 390.98, 213.73, 210.12, and 221.25 s for S0, SEQ, JFAST, and JSEL, respectively.

C. Sensitivity and computation

At load 0.60, halving the physical step changed the JSEL-minus-JFAST duration contrast by +0.30 [-0.06, 0.74] s; at 0.95 the change was -23.27 [-79.12, 7.76] s. The high-load estimate has substantially wider uncertainty. Disabling escalation increased JSEL duration by 4.32 [0.65, 8.92] s at load 0.60 and 38.82 [23.64, 55.36] s at 0.95. Spatial changes were -0.43 [-0.92, 0.01] and -17.65 [-49.90, 0.28] vehicle-hours.

With longest-count base signals, the duration contrast changed by −2.70 [−7.46, 2.05] s at load 0.60 and −8.07 [−31.39, 10.28] s at 0.95. At high load, the spatial contrast changed by +10.19 [−0.21, 22.84] vehicle-hours. More frequent noiseless observations changed the duration contrast by +4.25 [1.83, 6.81] s at 0.60 and −38.10 [−128.70, 12.08] s at 0.95. The latter spatial change was −18.21 [−56.57, 2.88] vehicle-hours. Each sensitivity uses 12 paired seeds per load. The observation comparison jointly varies sampling, noise, delay, and dropout timing.

Leaving out one seed at a time gave aggregate duration contrasts of +10.61 to +11.40 s, spatial contrasts of -4.04 to -3.67 vehicle-hours, and individual-residence contrasts of -1.72 to -1.53. These influence ranges show how each seed affects the balanced estimate. The main analysis retains every evaluation seed and reports bootstrap uncertainty separately.

Median simulator-loop time was 4.82 s per two-hour episode, with 95th percentile 7.54 s under the recorded concurrent workload. Planning and signal updates averaged 0.0308 ms per one-second tick; the 95th percentile of episode-average cost was 0.0423 ms. These amortized costs cover the simulator loop and controller updates, excluding input generation, network compilation, and trace export.

VIII. Discussion

A. Meaning of the measured trade-off

In the balanced microscopic comparison, JSEL reduced mean restricted ambulance duration by 44.6% relative to routing alone. It retained about 94% of the mean-time gain achieved by aggressive joint control, while reducing that controller's positive spatial exposure by about 5.8%. The paired contrasts quantify this exchange directly: 11.10 s of additional mean ambulance duration for 3.84 fewer vehicle-hours of spatial exposure. The engineering result is a measurable reduction in traffic disturbance while preserving most of the emergency-time benefit.

B. Mechanism of joint control

The controller coordinates three decisions: which route to follow, whether a local intervention merits its predicted traffic burden, and when observed mission progress warrants aggressive service. Joint route scoring incorporates anticipated priority into candidate travel times. Selective admission then combines estimated benefit, conflicting-approach burden, hold feasibility, and receiving space. These checks place the traffic displaced by a request within the decision itself. With $\varepsilon = 0$, burden ranking separates candidates tied at the minimum computed ETA, while selective admission governs which signal requests are executed.

Escalation complements this local selectivity through mission-specific feedback. It compares elapsed time with executed free-flow credit, switching control when measured progress deteriorates. Removing escalation increased mean duration by 4.32 s at load 0.60 and 38.82 s at 0.95. This ablation identifies a substantial contribution from progress feedback at high load. The 51.6% escalation rate describes the evaluated policy as an integrated hybrid: selective admission and stronger service jointly produce its measured performance.

C. What the matched simulations establish

The two traffic resolutions connect implementation mechanics with policy effects. Corrected mesoscopic reruns enforce source admission, finite storage, and PCU discharge accounting. Microscopic evaluation adds individual following, lane changes, junction connections, and overlapping missions. Across 1,728 held-out runs, matched scenario schedules and exact census-to-ledger reconciliation preserve the paired comparison. Spatial exposure records where occupancy increases accumulate; individual residence records traveler-level increases. Together, these measures explain how priority redistributes traffic and how that redistribution affects journeys, even when signed network residence combines gains and losses.

D. Modeling scope and applicability

The evaluation uses one synthetic district, fixed ordinary routes, nominal design loading, and finite-horizon accounting that retains source waiting and unfinished trips. Pedestrian service uses reserved signal time. Field application requires corridor-specific calibration of demand, incident behavior, emergency yielding, and lateral interactions. Seed-block intervals describe variability within the scenario generator; Section VII-B reports the associated mission-reliability outcomes.

IX. Conclusion

The controller combines finite route-action search, bounded selective priority, and progress escalation with common-cohort traffic accounting. Matched experiments connect the executable design to

ambulance duration, spatial exposure, and individual residence outcomes. Mechanical reanalysis establishes consistent admission, storage, and discharge accounting before policy effects are interpreted.

Held-out microscopic evaluation reduced balanced mean ambulance duration by 44.6% with JSEL relative to routing alone. Against aggressive joint control, JSEL added 11.10 s [9.47, 12.94] and reduced spatial exposure by 3.84 vehicle-hours [2.95, 4.73], satisfying the aggregate criterion. The contribution is an executable hybrid controller integrating route choice, intervention cost, and observed progress, supported by conservation checks and paired evaluation. It demonstrates how emergency-time gains and ordinary-traffic aftereffects can be measured and managed together.

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Appendix A
Demand and Nominal Signal Design

For each ordinary OD pair, up to four loop-free routes within 1.35 times the shortest free-flow time are assigned probabilities proportional to exp(−0.05 times excess free-flow seconds). Assigned ordinary routes remain fixed. The historical flows used by the controller follow from these public assignments.

Let f_e be assigned PCU flow under reference origin rate 0.1 vehicles/s, s_e saturation flow, and κ the common demand multiplier. At junction j with m_j approaches, define $y_e = \kappa\ f_e / s_e$ and $Y_j = \sum_{e \to j} y_e$. The Webster-inspired design [21] is

$$L_j = 25 + 7 m_j, \tag{35}$$

$$C_j^* = min\{150, max(60, c_j, 25 + 13 m_j)\}, \tag{36}$$

$$c_j = \frac{1.5 L_j + 5}{max(1 - Y_j, 0.05)}. \tag{37}$$

The 25 s term reserves pedestrian service; five seconds per approach cover amber and all-red, and two add a start-up allowance. Each approach receives an 8 s minimum green plus a demand-proportional share of the remaining green. Bisection selects κ so that the largest continuous-design ratio reaches the target:

$$X_e = \frac{\kappa\ f_e}{s_e (g_e / C_j^*)}. \tag{38}$$

Integer-rounded greens define the executable cycle. Nominal loading sets the design target; simulated trajectories characterize physical SUMO behavior.

Minute shocks follow $Z_m = 0.75 Z_{m-1} + \eta_m$, with independent Gaussian innovation standard deviation 0.18 and stationary initialization. Origin intensities are $0.1 \kappa \exp[Z_m - Var(Z_m)/2]$. Poisson release times are rounded upward to integer seconds. Separate namespaced streams generate destination, class, route, observations, and missions.

For M missions during the one-hour interval, mission $k = 0, \ldots, M-1$ is dispatched at

$$d_k = \lceil 1800 + \frac{3600}{M}(k + 0.15 + 0.70 U_k) \rceil, \quad (39)$$

where U is uniform on [0, 1). Origins are sampled from six boundaries and the destination is BE1. Jittered scheduled spacing distributes missions across the dispatch interval and permits overlap. Control receives missions when dispatched and traffic observations when available, preserving the experiment's causal information structure.

Appendix B
Executable Controller Procedure

The procedure specifies the frozen controller used for the reported runs. The reproducibility archive identifies its implementation and parameter selections. Scenario generation is managed by the simulator; the controller acts on dispatched missions and received observations. This separation makes the information available at each decision explicit.

Algorithm 1. One control update and route scoring

```
Receive available counts; hold stale observations.
For each active ambulance on a regular link:
  Update completed-link credit and current-link progress.
  If selective and progress deficit > γ:
    Set its irreversible flag; clear its lease; replan now.
  On link entry or escalation:
    Generate mode-specific route–action candidates.
    Score every candidate using Procedure SCORE.
    In JSEL, filter by the global minimum ETA + ε.
    Prefer burden, then ETA and deterministic identifiers.
    Apply 10 s incumbent preference only within that cap.
    In other modes, use the specified time-oriented choice.
    Install the continuation and retain its local action.
  Remove an old-link lease on regular-link departure.
  For a current selective lease:
    On first observed own green, set absolute expiry to
      now + max(0, requested hold − elapsed green).
    If expiry reached, admission age ≥ 120 s, or no
      receiving headroom: remove and mark nonrenewable.
    Otherwise forward the admitted request.
  Without a lease, apply trigger and admission gates;
    do not re-admit an expiry/headroom-finished pair.
  Use aggressive requests in aggressive modes.
Order conflicting requests by dispatch time and ID.
Execute service-age protection and legal stage changes.
Before each half-step, record the left-endpoint census;
  then advance and collect insertion/completion events.
Procedure SCORE(route, action, mode)
Initialize ETA and burden to zero.
For each regular link in the continuation:
  Add remaining/current or full/downstream free-flow
    time and the received-count discharge proxy.
  Set arrival offset τ to the accumulated ETA.
  If its downstream node is unsignalized: continue.
  Forecast residual nominal wait from the current stage.
  For S0/SEQ forecasts or action (0,0): add nominal
    wait and no burden; continue.
  Estimate legal-switch delay, feasibility, and headroom.
  Compute the local benefit and burden score.
  Admit benefit using the selected mode's gates.
  Add residual wait minus admitted benefit, clipped
    at zero, and add burden only if admitted.
Return accumulated ETA and admitted burden.
```

Selective scoring assigns zero intervention benefit and full nominal residual wait to requests that fail admission. Aggressive scoring uses its time-oriented mode, with receiving headroom as the scoring gate. Both use approximate forecasts from current signal and traffic observations. Processing an active selective lease before new benefit tests preserves an admitted request while successful green service reduces its estimated benefit. The execution record retains route-installation failures and physical simulation exceptions as explicit failure events.

The progress register counts actual regular-link traversals, including revisits. Connector time enters elapsed duration; regular links earn free-flow credit. The 20 s threshold diagnoses observed movement deficit and triggers aggressive control. Final journey effects are measured against the paired JFAST outcomes.

Appendix C
Mesoscopic Inputs and Reproduction Scope

The mesoscopic district shares the network geometry, class probabilities, and PCU factors stated in the main text. Saturation assumptions are 0.40–0.48 PCU/s/lane. Storage is the integer part of 0.16 times lane count times usable length after a 5 m stop-line buffer. The pooled-head approximation, unbounded exit-link transport, and abstract yielding are retained after mechanical correction.

One incoming approach is protected at a time, followed by 3 s amber and 2 s all-red. The pedestrian reservation comprises 7 s walk, 16 s clearance, and 2 s all-red. Discharge begins after 2 s start-up loss and continues through amber, giving effective green $g+1$ s under this model's convention. Webster-inspired nominal cycles are constrained to 60–150 s and green splits to 8–60 s. Integer timing gives maximum design load 0.79499 for target 0.80; other target deviations are below 0.0002. Guards and interventions subsequently change service, so this remains a nominal index.

Origin weights drawn uniformly from 230–390 vehicles/h are scaled together. A common stationary AR(1) lognormal multiplier has coefficient 0.75, log standard deviation 0.18, and unit expectation. Conditional releases are uniform within the minute. Poisson sampling is exact below mean 30, with a rounded-normal approximation above; the approximation was inactive in the evaluation, whose largest origin-minute mean was 14.608. Destination weights combine exp(−distance/2200 m) with attraction 1.6 for BE1, 1.2 for BE2 and BS1, and 1 otherwise. At most four ordinary routes within a free-flow factor of 1.25 use logit sensitivity 0.05 per second.

An incident occurs with probability 0.5 on a random two-lane non-exit link, starts from 180 s before to 120 s after dispatch, and lasts 240–480 s. One lane retains 45–65% discharge capacity, giving aggregate multiplier 0.725–0.825. Speed multipliers 0.75–0.90 affect new link-entry readiness clocks; previously assigned clocks are retained.

Camera samples arrive every 5 s with zero additional receipt delay. Count noise and per-sample availability transitions follow the probabilities in the main text. With probability 0.5, a further 180-420 s outage affects approaches within 800 m Manhattan distance of a sampled junction. Abstract yielding reorders vehicles ahead of an ambulance with class probabilities y times (0.6, 1.0, 0.8, 0.2, 0.2), conserving occupancy within the pooled-queue model. Policy-dependent encounters consume yielding draws sequentially, making each realized yielding sequence dependent on the resulting vehicle interactions.

M1 extends green by at most 10 s or truncates to the 8 s minimum, retaining phase order. M2 permits phase skipping after clearance and holds up to 60 s with 30 s look-ahead. MQ uses the inherited 15 s queue-aware trigger, 10 s margin, and 90 s hold cap. MC compensates nominal service-share deficits within guard slack; MR adds link-entry rerouting when estimated improvement exceeds 10 s. Started pedestrian reservations complete in full. All policies use a stage-selection deadline guard; priority modes add per-tick protection during priority and for two nominal cycles after release. These rules protect service opportunities within the mesoscopic signal model.

Evaluation uses nominal loads 0.60, 0.70, 0.80, 0.85, 0.90, and 0.95; yielding parameters 0 and 0.5; and 200 seeds per condition. One ambulance departs from six possible boundary origins at integer time 1,800-2,100 s and heads to BE1. Episodes end at 4,200 s. Retained tuning uses seeds 5000-5039 at loads 0.80 and 0.90 under both yielding settings. Corrected reruns preserve those selections, yielding a matched sensitivity analysis of the mechanical changes.

Appendix D
Full Microscopic Results

Each table cell uses 48 paired scenario seeds, with values computed from the retained analysis. Ordinary-traffic outcomes use the matching S0 episode as reference. Reported intervals are marginal seed-block intervals for the stated contrast, retaining the same pairing and accounting definitions as the main analysis.

TABLE III
Cell means at nominal load 0.60

Dose	Policy	Time (s)	A (veh-h)	D^+ (veh-h)	ΔJ (veh-h)
1	S0	353.07	0.00	0.00	0.00
1	SEQ	199.49	23.51	6.15	-0.90
1	JFAST	195.35	24.29	6.35	-1.13
1	JSEL	203.05	20.63	5.34	-1.14
4	S0	367.32	0.00	0.00	0.00
4	SEQ	206.39	50.45	13.02	-1.67
4	JFAST	200.57	48.75	12.50	-2.24
4	JSEL	208.36	46.19	11.76	-2.39
10	S0	367.26	0.00	0.00	0.00
10	SEQ	208.60	65.68	16.99	-0.98
10	JFAST	201.67	63.81	16.64	-1.08
10	JSEL	209.63	62.01	15.81	-2.03

TABLE IV
Cell means at nominal load 0.80

Dose	Policy	Time (s)	A (veh-h)	D^+ (veh-h)	ΔJ (veh-h)
1	S0	378.51	0.00	0.00	0.00
1	SEQ	202.08	31.33	10.32	0.17
1	JFAST	208.45	32.84	10.66	-0.04
1	JSEL	217.71	29.91	9.73	-0.33
4	S0	397.04	0.00	0.00	0.00
4	SEQ	214.49	68.16	22.16	2.25
4	JFAST	211.11	68.07	22.14	2.48
4	JSEL	221.20	65.20	21.44	2.33
10	S0	395.17	0.00	0.00	0.00
10	SEQ	215.43	99.42	32.66	6.34
10	JFAST	210.80	99.89	33.44	7.78
10	JSEL	222.01	95.42	31.51	6.29

TABLE V
Cell means at nominal load 0.95

Dose	Policy	Time (s)	A (veh-h)	D^+ (veh-h)	ΔJ (veh-h)
1	S0	430.64	0.00	0.00	0.00
1	SEQ	219.34	34.75	16.31	1.51
1	JFAST	217.34	35.91	16.84	1.72
1	JSEL	235.15	34.63	16.28	1.63
4	S0	461.73	0.00	0.00	0.00
4	SEQ	231.92	91.60	38.44	4.63
4	JFAST	226.56	93.08	40.34	9.24
4	JSEL	242.73	84.50	36.75	6.65
10	S0	463.41	0.00	0.00	0.00
10	SEQ	234.35	128.76	50.71	5.41
10	JFAST	228.81	131.86	56.47	18.30
10	JSEL	240.76	125.40	52.00	12.62

TABLE VI
JSEL minus JFAST: time and spatial exposure. Mean [95% CI].

Load / dose	Time (s)	A (veh-h)
0.60 / 1	7.70 [5.44, 10.11]	-3.65 [-4.92, -2.44]
0.60 / 4	7.78 [6.43, 9.14]	-2.56 [-3.90, -1.34]
0.60 / 10	7.96 [6.91, 9.03]	-1.80 [-2.77, -0.81]
0.80 / 1	9.26 [3.62, 15.47]	-2.93 [-4.73, -1.22]
0.80 / 4	10.09 [7.17, 13.01]	-2.87 [-4.91, -0.92]
0.80 / 10	11.21 [9.22, 13.46]	-4.47 [-6.55, -2.36]
0.95 / 1	17.80 [9.74, 27.17]	-1.28 [-3.04, 0.52]
0.95 / 4	16.17 [10.83, 22.84]	-8.58 [-12.90, -4.66]
0.95 / 10	11.95 [6.14, 16.34]	-6.45 [-10.13, -2.81]

TABLE VII
JSEL minus JFAST: individual and signed residence. Mean [95% CI].

Load / dose	D^+ (veh-h)	ΔJ (veh-h)
0.60 / 1	-1.01 [-1.41, -0.63]	-0.01 [-0.44, 0.42]
0.60 / 4	-0.74 [-1.15, -0.35]	-0.15 [-0.53, 0.23]
0.60 / 10	-0.83 [-1.21, -0.43]	-0.94 [-1.44, -0.47]
0.80 / 1	-0.94 [-1.59, -0.31]	-0.29 [-0.85, 0.17]
0.80 / 4	-0.69 [-1.61, 0.18]	-0.15 [-1.08, 0.87]
0.80 / 10	-1.92 [-2.78, -1.11]	-1.49 [-2.57, -0.45]
0.95 / 1	-0.56 [-1.42, 0.25]	-0.08 [-1.10, 1.02]
0.95 / 4	-3.59 [-6.53, -1.23]	-2.58 [-5.75, 0.24]
0.95 / 10	-4.47 [-7.07, -1.97]	-5.68 [-10.02, -1.81]

For Tables VIII–X, "No escalation" is unguarded minus guarded JSEL. Other rows give the change in the JSEL-minus-JFAST effect. Each load uses 12 paired seeds.

TABLE VIII
Sensitivity in duration (s)

Change	Load	Mean [95% CI]
0.25 s step	0.60	0.30 [-0.06, 0.74]
0.25 s step	0.95	-23.27 [-79.12, 7.76]
No escalation	0.60	4.32 [0.65, 8.92]
No escalation	0.95	38.82 [23.64, 55.36]
Longest count	0.60	-2.70 [-7.46, 2.05]
Longest count	0.95	-8.07 [-31.39, 10.28]
Frequent counts	0.60	4.25 [1.83, 6.81]
Frequent counts	0.95	-38.10 [-128.70, 12.08]

TABLE IX
Sensitivity in spatial exposure (vehicle-hours)

Change	Load	Mean [95% CI]
0.25 s step	0.60	-0.25 [-2.30, 1.41]
0.25 s step	0.95	-1.45 [-7.12, 5.10]
No escalation	0.60	-0.43 [-0.92, 0.01]
No escalation	0.95	-17.65 [-49.90, 0.28]
Longest count	0.60	1.35 [-2.57, 5.34]
Longest count	0.95	10.19 [-0.21, 22.84]
Frequent counts	0.60	-0.92 [-4.58, 2.23]
Frequent counts	0.95	-18.21 [-56.57, 2.88]

TABLE X
Sensitivity in positive individual residence (vehicle-hours)

Change	Load	Mean [95% CI]
0.25 s step	0.60	-0.03 [-0.67, 0.48]
0.25 s step	0.95	-2.68 [-7.02, 0.36]
No escalation	0.60	-0.02 [-0.17, 0.17]
No escalation	0.95	-12.80 [-38.31, 1.01]
Longest count	0.60	-0.18 [-1.72, 1.40]
Longest count	0.95	3.69 [-3.73, 13.46]
Frequent counts	0.60	0.14 [-0.85, 1.19]
Frequent counts	0.95	-16.77 [-51.56, 2.50]

Rare-event uncertainty uses independent episodes within each fixed load-dose condition. The event is at least one mission exceeding 600 s, including noncompletion, among 48 episodes per cell. Exact binomial intervals quantify uncertainty at that episode level. Mission-fraction bootstrap summaries are reported separately. The accompanying computational records contain all policies; Table XI presents the JSEL episode results.

TABLE XI
JSEL episodes with at least one mission above 600 s

Load / dose	Late episodes	Exact 95% interval
0.60 / 1	0/48	0.000–7.397%
0.60 / 4	0/48	0.000–7.397%
0.60 / 10	0/48	0.000–7.397%
0.80 / 1	0/48	0.000–7.397%
0.80 / 4	0/48	0.000–7.397%
0.80 / 10	0/48	0.000–7.397%
0.95 / 1	0/48	0.000–7.397%
0.95 / 4	1/48	0.053–11.070%
0.95 / 10	1/48	0.053–11.070%

Appendix E
Reproducibility

The retained computational package contains original and corrected mesoscopic sources, frozen microscopic implementations, input fixtures, policy selections, protocols, episode summaries, population traces, individual event records, and statistical scripts. Development and held-out evaluation runs are identified separately, and execution exceptions stop the run. The frozen microscopic source is identified by SHA-256: 471a22911a2de9773fcd839872b3ba1ec5e00412c0ecc79142503e72f41cecf2. This identifier connects the reported experiments to the exact retained implementation.

Population histories and ordinary event records support reconstruction of the residence identities. Each intervention policy uses the matching S0 history in its scenario group. Physical-step sensitivity fixes integer-second desired releases and one-second control decisions. SUMO's default vehicle action interval follows the physical step, allowing lane order and trajectories to change with driver timing. The empirical results correspond to the frozen implementations; the equations and pseudocode describe the same evaluated procedures and support reproduction of their control and accounting logic.